MAT 3007 Homework # 4 Spring 2024

Due: Wednesday, March 20 at 12:00pm (noon)

Please note: You should submit your answers (scanned or typeset) in a PDF file. Question 2 asks you to solve a model in Python or MATLAB. Submit all necessary code files via Blackboard. Also, give meaningful names to your decision variables and constraints, and add comments to your code liberally. The problem set is due at 12:00pm (noon) on Wednesday, March 20th.

1. (15 pts) Consider the following linear program:

$$\begin{aligned} \max \quad & 5x_1 + 2x_2 + 5x_3 \\ \text{s.t.} \quad & 2x_1 + 3x_2 + x_3 \leq 4 \\ & x_1 + 2x_2 + 3x_3 \leq 7 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

(a) Write the formulation for the corresponding dual problem.

(b) Solve the dual problem in part (a) graphically.

(c) Use complementarity slackness to solve the primal problem.

2. (20 pts) Consider the following table of food and its nutritional values:

	Protein, g	Carbohydrates, g	Calories	Cost
Bread	4	8	130	2
Milk	6	12	120	3.5
Fish	20	0	150	8
Potato	1	30	70	1.5

The ideal intake for an adult is at least 25 grams of protein, 40 grams of carbohydrates, and 400 calories per day. The problem is to find the least costly way to achieve those amounts of nutrition by using the four types of food shown in the table.

(a) Formulate this problem as a linear optimization problem (specify the meaning of each decision variable and constraint).

(b) Solve it using MATLAB or Python, find the optimal solution and the optimal value.

(c) Formulate the dual problem. Interpret the dual problem in a way a person without any optimization background can understand. (Hint: Suppose a pharmaceutical company produces each of the nutrients in pill form and sells them each for a certain price.)

(d) Use MATLAB or Python to solve the dual problem. Find the optimal solution and the optimal value.

3. (20 pts) Use linear program duality theory to show that exactly one of the following systems has a solution

(a) $A\boldsymbol{x} \leq \boldsymbol{b}$

(b) $\boldsymbol{y}^T A = 0$, $\boldsymbol{b}^T \boldsymbol{y} < 0$, $\boldsymbol{y} \geq 0$

Hint: You can first show that it can't both have solutions. Then you show that if the second one is infeasible, the first one must be feasible.

4. (20 pts) A manager of an oil refinery has 8000 barrels of crude oil A and 5000 barrels of crude oil B allocated for production during the coming month. These resources can be used to make either gasoline, which sells for $38 per barrel, or home heating oil, which sells for $33 per barrel. There are three production processes with the characteristics specified in the following table (all quantities are in barrels):

	Process 1	Process 2	Process 3
Input Crude A	3	1	5
Input Crude B	5	1	3
Output Gasoline	4	1	3
Output Heating Oil	3	1	4
Cost	$51	$11	$40

(a) First formulate a linear optimization problem that would help the manager maximize net revenue over the next month. Then use the simplex method to find an optimal solution. Please write down each iteration.

(b) Suppose that the selling price of heating oil remains fixed over the next month, but the selling price of gasoline may rise. Using the simplex tableau, try to decide how high it can go without causing the optimal solution to change.

(c) Now the manager has to consider waste disposal as well. Suppose per-unit Process 1 generates 4 tons of waste and per-unit Process 2 and 3 generate 3 tons and 5 tons of waste respectively. The refinery can dispose of 10000 tons of waste every month. What will be the optimal solution with this constraint?

5. (25 pts) An insurance company is introducing three products: special risk insurance, mortgage insurance, and long-term care insurance. The expected profit is \$500 per unit on special risk insurance, \$250 per unit on mortgage insurance and \$600 per unit on long term care insurance. The work requirements are as follows:

Department	Working hours per unit			Working hours available
	Special risk	Mortgage	Long-term care	
Underwriting	2	1	1	240
Administration	3	1	2	150
Claims	1	2	4	180

The management team wants to establish sales quotas for each product to maximize the total expected profit.

(a) Formulate this problem as a linear optimization problem. Specify the decision variables, objective function, and constraints.

(b) After solving the problem, the final simplex tableau (for the standard form) is given as below (the variables are in the natural order as in the description of the problem):

B	0	50	0	0	140	80	35400
4	0	0.5	0	1	-0.7	0.1	153
1	1	0	0	0	0.4	-0.2	24
3	0	0.5	1	0	-0.1	0.3	39

Show the dual variables corresponding to the services of the three departments. Using complementarity conditions to explain why mortgage insurance is not sold.

(c) Find the range of working hours available for claims to keep the current basis optimal.

(d) Find the range of the expected profit on special risk insurance such that the current basis remains optimal.